Mat 1341A Test 4 v1

1. Let A be a 4×5 matrix. Recall that $\ker A = \{x \in \mathbf{R}^5 \mid Ax = 0\}$ and $\operatorname{col} A = \{Ax \mid x \in \mathbf{R}^5\}$. If $\operatorname{rank} A = 4$, then

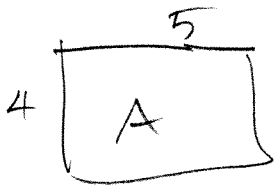
A. $\operatorname{col} A = \mathbf{R}^4$ and $\dim \ker A = 1$
B. $\operatorname{col} A = \mathbf{R}^4$ and $\dim \ker A = 4$
C. $\dim \operatorname{col} A = 1$ and $\dim \ker A = 1$
D. $\dim \operatorname{col} A = 1$ and $\ker A = \mathbf{R}^5$
E. $\dim \operatorname{col} A = 4$ and $\ker A = \mathbf{R}^5$
F. $\operatorname{col} A = \{0\}$ and $\dim \ker A = 4$

rank A + dim ker A = 5

∴ dim ker A = 5 − 4 = 1

& dim col A = rank A = 4

Since col A ⊆ $\mathbb{R}^4$, which has dimension 4, col A = $\mathbb{R}^4$.



2. Let $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$. The third row of B^{-1} is:

A. $[0 \;\; 1 \;\; -1]$
B. $[-1 \;\; 1 \;\; 0]$
C. $[0 \;\; -1 \;\; 1]$
D. $[1 \;\; -1 \;\; 0]$
E. $[1 \;\; 0 \;\; -1]$
F. B is not invertible.

$$(B|I_3) = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & -1 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & & * & \\ 0 & 1 & -1 & & * & \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right]$$

V1

3. Let $S = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and consider the subset $W = \{A \in \mathbf{M}_{22} \mid SA = AS\}$. Which one of the following statements is true?

A. W is not a subspace of $\mathbf{M}_{22}$
B. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 4$
C. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 3$
D. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 2$
E. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 1$
F. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 0$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $AS = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & a \\ c & c \end{bmatrix}$ solve

$SA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix} \therefore SA = AS$

$\Leftrightarrow c = 0$ and $a = b + d$

$\therefore W = \left\{ \begin{bmatrix} b+d & b \\ c & d \end{bmatrix} \middle| b, d \in \mathbb{R} \right\}$

$= \operatorname{span}\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

l.i.

$\therefore \dim W = 2$

4. Let

$$A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 2 & 2 & -1 & 1 \\ 1 & 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 3 \\ -2 & 0 & -1 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (*)$$

a) Find a basis for the row space of A. (row space alg.)

b) Find a basis for the column space of A. (col. space alg.)

c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

d) Find the dot product of each of the vectors in your basis from (a) with each of the vectors in your basis in (c).

a) From (*), $\{\underbrace{(1,1,0,-1)}_{u_1}, \underbrace{(0,0,1,-3)}_{u_2}\}$ is a basis of row(A).

---

b) From (*) columns 1 & 3 of A will be a basis of col A, namely $\{(1,2,1), (0,-1,-1)\}$.

---

c) From (*) $x_1 = -s + t$, $x_2 = s$, $x_3 = 3t$, $x_4 = t$; $s,t \in \mathbb{R}$

$\therefore \{\underbrace{(-1,1,0,0)}_{u_3}, \underbrace{(1,0,3,1)}_{u_4}\}$ is a basis of $\ker A$

---

d) $u_1 \cdot u_3 = -1 + 1 = 0$

$u_1 \cdot u_4 = 1 - 1 = 0$

$u_2 \cdot u_3 = 0 + 0 = 0$

$u_2 \cdot u_4 = 3 - 3 = 0$

5. Let $U = \text{span}\{\underset{u_1}{(1,0,0,1)}, \underset{u_2}{(1,0,0,0)}, \underset{u_3}{(0,1,-1,0)}\}$.

a) Use the Gram Schmidt algorithm to find an orthogonal basis of U.

b) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

c) Extend your basis in (a) to a basis of $\mathbf{R}^4$.

a) Set $w_1 = u_1$

$$w_2 = u_2 - \frac{u_2 \cdot w_1}{\|w_1\|^2} w_1 = (1,0,0,0) - \frac{1}{2}(1,0,0,1)$$

$$= \left(\tfrac{1}{2}, 0, 0, -\tfrac{1}{2}\right). \quad \text{Use } \tilde{w}_2 = (1,0,0,-1)$$

$$w_3 = u_3 - \frac{u_3 \cdot w_1}{\|w_1\|^2} w_1 - \frac{u_3 \cdot \tilde{w}_2}{\|\tilde{w}_2\|^2} \tilde{w}_2$$

$$= (0,1,-1,0) - 0 w_1 - 0 = (0,1,-1,0)$$

$\therefore \{w_1, \tilde{w}_2, w_3\} = \{(1,0,0,1), (1,0,0,-1), (0,1,-1,0)\}$ is an orthogonal basis for U.

b) $\text{proj}_U (1,-1,2,-1) = \dfrac{(1,-1,2,-1)\cdot(1,0,0,1)}{2}(1,0,0,1) + \dfrac{(1,-1,2,-1)\cdot(1,0,0,-1)}{2}(1,0,0,-1)$

$$+ \frac{(1,-1,2,-1)\cdot(0,1,-1,0)}{2} \cdot (0,1,-1,0)$$

$$= 0 \cdot w_1 + \frac{2}{2}(1,0,0,-1) - \frac{3}{2}(0,1,-1,0)$$

$$= \left(1, -\tfrac{3}{2}, \tfrac{3}{2}, -1\right) \text{ is the required answer}$$

c) $\begin{bmatrix} w_1 \\ \tilde{w}_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\therefore \{w_1, \tilde{w}_2, w_3, e_3\}$

is a suitable extension to a basis of $\mathbb{R}^4$

√1

6(a). State whether the following is true or false, and justify your answer.

(i) If A is an invertible matrix and $AB = 0$ then $B = 0$.

Suppose $AB = 0$. Then $A^{-1}(AB) = A^{-1}(0) = 0$

But $A^{-1}(AB) = (A^{-1}A)B = B$. So $B = 0$

ANSWER TRUE

(ii) The columns of a 4 by 3 matrix are always linearly dependent.

eg Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Then the cols of A are independent.

ANSWER FALSE

√1

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"The system $Ax = 0$ has infinitely many solutions" (So A is not invertible

(i) In terms of the rank of A:

$\text{rank} A < n$

(ii) In terms of the reduced row echelon form of A: It has a row of zeros

(iii) In terms of the columns of A: They are linearly dependent